

Koi-Friend: IoT-Enabled Aquaculture System with Feeding, Mortality Detection, Budgeting, Monitoring, and Alerts

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Abstract—Koi fish farming requires continuous monitoring of feeding, fish health, water quality, and operational cost, as ornamental fish are highly sensitive to environmental and management-related changes. Conventional koi farming practices mainly depend on manual feeding, periodic water testing, and visual observation. These approaches may lead to inconsistent feeding, delayed identification of unhealthy or dead fish, unstable pond conditions, and limited financial visibility. This paper presents Koi-Friend, an IoT-enabled smart aquaculture system developed to support koi pond management. The proposed system integrates four main modules: fixed automated feeding, fish mortality detection and environmental alerting, smart maintenance with water quality monitoring, and budget tracking with predictive forecasting. The feeding module uses an ESP32-based fixed device with load cells, motor control, solar power, and cloud-based scheduling to dispense accurate feed quantities. The mortality detection module uses Raspberry Pi-based live video processing with computer vision and behaviour-based analysis to identify abnormal fish conditions over time. The water quality module monitors pH, turbidity, temperature, dissolved oxygen, ammonia, carbon dioxide, and rainfall, while generating condition-based maintenance reminders. The budget module combines expense recording, energy monitoring, alerts, and time-series forecasting to support financial planning. The proposed system enables real-time monitoring, early risk detection, improved cost visibility, and practical decision support for small and medium-scale koi farmers.

Keywords— Automated Feeding, Cost Forecasting, Fish Mortality Detection, Internet of Things, Smart Aquaculture, Water Quality Monitoring

I. INTRODUCTION

Aquaculture contributes significantly to food production, rural livelihoods, and the ornamental fish industry, particularly where small and medium-scale farms support local income generation [1]. Koi farming is a specialized ornamental sector in which fish health, color quality, growth, and survival affect economic value. Koi fish are sensitive to feed quantity, water quality, in pond conditions may increase stress, support disease development, and cause mortality. Therefore, koi pond management requires continuous

monitoring, accurate feeding, early risk identification, and cost control.

Traditional koi farming mainly depends on manual feeding, periodic water testing, and visual observation of fish behavior [2]. Although widely used, these methods are labor intensive, inconsistent, and dependent on the farmer's experience and availability. Manual feeding may cause underfeeding, overfeeding, uneven distribution, and feed wastage. Excess feed can reduce water quality and increase production cost. Delayed identification of unhealthy or dead fish may also allow minor issues to develop into serious losses. In many farms, feeding records, water readings, energy usage, maintenance activities, and financial data are managed separately, limiting timely decisions.

Recent developments in the Internet of Things (IOT), edge computing, computer vision, and cloud platforms provide practical opportunities for aquaculture management [3]. IOT devices support continuous sensing, automated control, and remote monitoring, while computer vision can identify abnormal fish conditions from live video streams. Cloud services and web dashboards support data storage, alerts, reporting, and decision support. However, many existing smart aquaculture solutions focus on individual tasks, such as water quality monitoring or feeding automation. Integrated systems that combine feeding, fish mortality detection, water quality monitoring, maintenance support, energy monitoring, and cost forecasting for koi farming remain limited.

This paper presents Koi-Friend, and IoT enables smart aquaculture systems for koi pond management. The system integrates a fixed automated feeding device, fish mortality detection and environmental alerting, smart maintenance with water quality monitoring, and budget tracking with predictive forecasting. By combining sensing, automation, edge-based analysis, real time communication, and financial analytics in a we based platform, the system aims to reduce manual effort, minimize fish loss, and support sustainable decision making for koi farmers.[4].

II. LITERATURE REVIEW

The application of IoT technology in aquaculture has received considerable attention due to the need for continuous monitoring, automation, and timely decision making in pond environments. Previous studies have shown that IoT based sensor networks can be used to monitor water quality parameters such as pH, temperature, dissolved oxygen, and turbidity in real time [5]. These systems reduce the limitations of manual testing and allow farmers to respond more quickly to environmental changes. However, many existing implementations focus mainly on water quality monitoring and do not fully integrate feeding automation, fish health assessment, maintenance support, and financial decision making within a single platform.

Feeding management is one of the most important operational activities in aquaculture because it directly affects fish growth, water quality, and production cost. Inefficient feeding practices may cause feed wastage, poor growth performance, and deterioration of pond conditions [6]. Automated feeding systems have been introduced to improve feeding regularity and reduce manual labor. However, many available systems are limited to basic timer-based feeding and do not include accurate feed measurement, remote scheduling, energy monitoring, or integration with cost analytics. This created a need for feeding systems that can provide controlled dispensing, real time logging, and connection with broader farm management functions.

Computer vision and deep learning have also become important in smart aquaculture research. Object detection models, particularly the YOLO family, are widely used because of their real-time performance and suitability for visual monitoring tasks [7]. These models have been applied in fish detection, counting, and behavior analysis with promising results [8]. YOLOv8 further improves flexibility for deployment in practical systems, including edge devices with limited computing resources [9]. Nevertheless, single-frame visual classification alone may not be sufficient for reliable mortality detection, since fish condition can vary due to movement, lighting, water reflection, and camera angle. Therefore, mortality detection can be improved by combining image-based detection with temporal behavior analysis, such as inactivity and abnormal floating patterns.

Sustainable energy management is another important requirement for outdoor aquaculture systems. Solar-powered IoT solutions have been proposed as practical alternatives for farms with limited access to stable grid electricity [10]. In pond-based deployments, power management becomes more important because sensors, motors, communication modules, and edge processors may operate continuously. Existing studies often discuss energy harvesting, but fewer systems include priority-based power control, battery monitoring, safe shutdown logic, and module-level energy awareness. These functions are essential for maintaining system reliability in remote or outdoor aquaculture environments.

Financial monitoring and cost forecasting remain less developed areas in smart aquaculture literature. Although IoT systems can collect operational data, farmers often lack tools

to convert these data into financial insights. Time-series forecasting methods, such as ARIMA, have been widely used for predicting future values in cost and resource planning [11]. When combined with expense records and energy consumption data, forecasting can support budget planning, cost control, and early identification of abnormal spending patterns.

The reviewed literature confirms the value of IoT sensing, automated feeding, computer vision, solar energy, and forecasting techniques in aquaculture. However, existing solutions are often fragmented and focused on individual problems. A unified system that combines fixed automated feeding, behavior-supported mortality detection, real-time water quality monitoring, condition-based maintenance reminders, energy monitoring, alerts, and predictive budgeting for koi farming remains limited. This study addresses that gap by proposing an integrated IoT-enabled aquaculture platform designed specifically for practical koi pond management.

III. METHODOLOGY

The below figure 1 represents the overall system architecture of the proposed IoT-enabled aquaculture system.

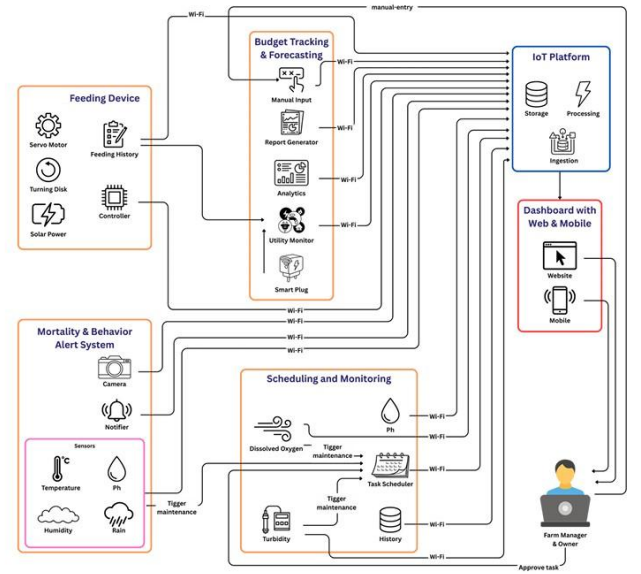


Fig. 1. Overall system architecture

The proposed system was developed using a modular IoT-based design approach that integrates sensing, automation, edge processing, cloud communication, and web-based visualization. As shown in Fig. 1, the system consists of four main functional modules: fixed automated feeding, fish mortality detection and environmental alerting, smart maintenance with water quality monitoring, and budget tracking with predictive forecasting. Each module was designed as an independent unit and then integrated through a centralized backend, MQTT communication, real-time data storage, and a React-based dashboard to support monitoring, control, alerts, reporting, and decision-making for koi pond management.

A. Automated Feeding System

The automated feeding system was implemented as a fixed IoT-based device installed near the pond. This fixed design improves mechanical stability and allows the feeding process to operate independently from the floating monitoring unit. The device uses an ESP32 microcontroller for control and communication, while load cells connected through an HX711 module measure the feed quantity during dispensing. A motor-driven mechanism releases feed according to the feeding schedule and quantity configured through the web dashboard. Manual feeding can also be activated remotely when required.

The feeding unit is powered by a solar panel and a 12 V battery, while a voltage sensor monitors the battery condition during operation. Each feeding event records the feeding time, feed quantity, device status, and battery level. These data are sent to the backend through MQTT communication and stored for monitoring and analysis. Alerts are generated for low battery level, empty feed storage, motor malfunction, or missed scheduled feeding. The recorded feed usage is also connected with the budget module to support feed cost calculation and cycle-based cost analysis.

B. Smart Floating Boat and Power Management

A smart floating boat was used to carry the fish health monitoring and water quality sensing modules across the pond. The boat is driven by two DC motors and controlled remotely through the web dashboard. Four ultrasonic sensors are mounted on the boat to detect obstacles and reduce collision risk during navigation.

The boat is powered by a solar panel and a rechargeable battery. Voltage and current sensors monitor the battery condition in real time. When the battery level decreases, the system gives priority to essential navigation functions and disables non-essential modules. At a critical battery level, the system shuts down safely and resumes operation after sufficient solar charging.

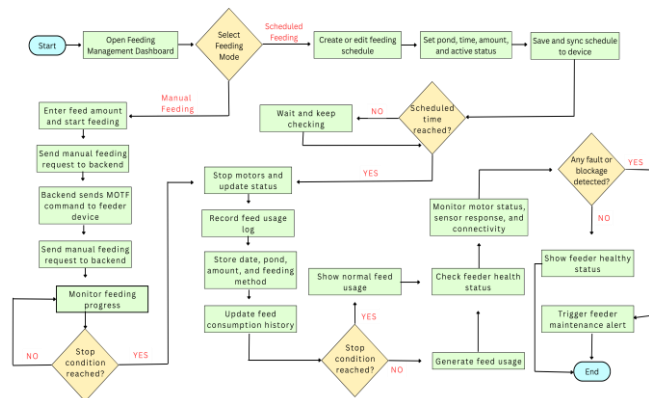


Fig. 2. Automated Feeding System Flow Chart

C. Fish Health and Mortality Detection

Fish health monitoring was carried out using a Raspberry Pi and camera module mounted on the floating platform. The system captures live video from the pond and processes the

frames in real time. A YOLOv8-based computer vision model is used to detect fish, while fish condition is analyzed using image-based features and behavior patterns such as low movement and side-floating posture.

A self-curated dataset of approximately 5,000 fish images was used for model training and validation. The trained model was deployed on the Raspberry Pi to support local inference, reduce latency, and minimize dependency on continuous internet connectivity. Detection results are sent to the backend and displayed on the web dashboard. When unhealthy or dead fish conditions are identified, alerts are generated through the dashboard and SMS notifications to support timely action.

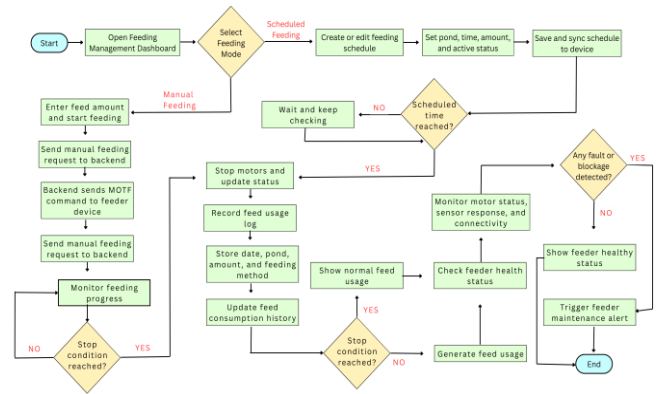


Fig. 3. Live Fish Health Monitoring Flow Chart

D. Water Quality Monitoring

Water quality monitoring was implemented using IoT sensors to measure turbidity, pH, ammonia, dissolved oxygen, carbon dioxide, and temperature. Since some sensors are not suitable for continuous direct submersion, they are mounted on the floating unit above the water surface, and a small pump draws pond water into a sensing chamber for measurement.

A rain sensor is also included to detect rainfall events that may affect water quality and fish behaviour. Sensor readings are sent to the backend at regular intervals and displayed on the web dashboard. Farmers can configure safe threshold levels for each parameter. When readings exceed the defined limits, the system generates alerts and automatically creates maintenance reminders to support proactive pond management.

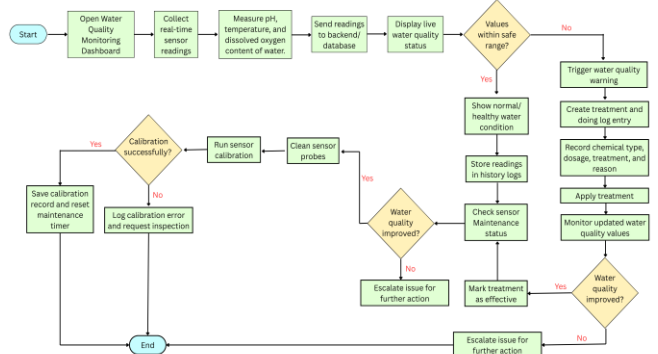


Fig. 4. Water Quality Monitoring Flow Chart

E. Budget Tracking and Forecasting

The budget tracking module combines manual expense recording with IoT-based energy monitoring. A POS-style interface allows farmers to record costs related to feed, labour, chemicals, maintenance, and utilities. Voltage and current sensors measure electricity usage, while relay-based control supports manual, scheduled, and safety-based power management.

The collected financial and energy data are analyzed to calculate key indicators such as total production cost, cost per farming cycle, and cost per surviving fish. Time-series forecasting methods are applied to historical cost data to estimate future expenses and support financial planning. Alerts are generated when abnormal energy usage, budget deviations, or unsafe voltage conditions are detected. Cost summaries, trends, forecasts, and reports are displayed through the web dashboard.

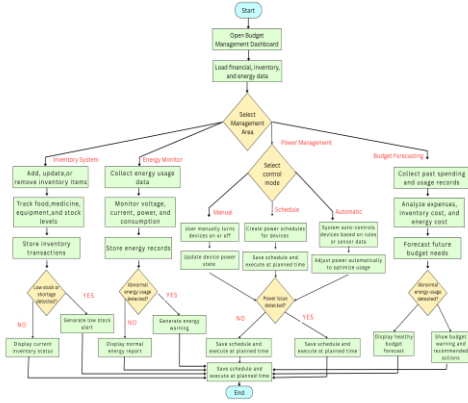


Fig. 5. Budget Management Flow Chart

F. System Integration and Data Flow

All system modules communicate with the backend through Spring Boot REST APIs and EMQX-based MQTT messaging. The backend manages data storage, real-time updates, alert processing, and communication between hardware devices and the web application. Supabase and SQL-based storage are used to maintain sensor readings, feeding logs, detection results, maintenance records, alerts, and budget data.

The React.js web dashboard provides real-time visualization, manual control, feeding schedules, live video streaming through MediaMTX, alert notifications, reports, and forecasting outputs. This modular integration allows each component to operate independently while sharing data through a centralized platform. Therefore, the system can be extended in the future without changing the overall architecture.

TABLE 1. Major System Component and Their Functions

Component	Hardware/ Software Used	Function
Automated Feeding System	NEMA 17 motor, 5 kg load cell, HX711,	solar power Precise feed dispensing and uniform distribution
Fish Health Monitoring	Raspberry Pi, camera, YOLOv8 models	Live detection of healthy, unhealthy, and dead fish
Water Quality Monitoring	pH, turbidity, NH ₃ , CO ₂ , DO sensors	Real-time water parameter monitoring and alerts
Budget and Forecasting	POS module, energy data, time-series analysis	Cost tracking and financial prediction
Smart Boat and Power Management	DC motors, ultrasonic sensors, solar panel, Voltage/current sensors	Mobile sensing, navigation, and obstacle avoidance, control relays Energy prioritization and safe shutdown
Web Dashboard	React.js, Spring Boot, EMQX, MediaMTX	Monitoring, control, alerts, and reporting

G. Implementation Environment and Tools

The proposed system was implemented using IoT hardware platforms, edge computing devices, and web technologies to support affordability, scalability, and practical deployment in koi fish farming environments. The web dashboard was developed using React.js to provide an interactive interface for monitoring, control, alert handling, reporting, and visualization. Spring Boot was used to develop the backend services and REST APIs required for communication between the system modules and the dashboard. MySQL was used as the primary database for storing system records, sensor readings, feeding logs, detection results, alerts, and budget-related data, while Supabase was used to support selected backend services and real-time updates.

For IoT communication, EMQX was used as the MQTT broker to enable lightweight and reliable message exchange between distributed devices and backend services. MediaMTX was used to manage live video streaming from the pond monitoring camera to the web dashboard. ESP32 microcontrollers were used for sensor interfacing, actuator control, energy monitoring, and embedded device communication. A Raspberry Pi was used as the edge processing device for the fish mortality detection module, where YOLOv8-based inference and video processing were executed locally to reduce latency and minimize network dependency.

The overall system was deployed on a Hostinger KVM 2 VPS using Docker-based containerized services. This deployment approach supports flexible service management, easier maintenance, and improved scalability. Communication among components was achieved using both REST APIs and MQTT protocols to support modular integration and real-time system operation [12].



Fig. 6. IoT Devices and Hardware Prototypes

H. IoT Devices and Hardware Prototypes Implemented

This research involved the design and implementation of several IoT hardware prototypes to support automated feeding, fish health monitoring, water quality analysis, energy monitoring, and system control. Each device was designed for outdoor pond environments with consideration for energy efficiency, reliability, and modular integration.

The fixed automated feeding device was installed near the pond as a stationary unit. It uses an ESP32 microcontroller for control and communication, load cells connected through an HX711 module for feed weight measurement, and a motor-driven dispensing mechanism to release feed according to the configured schedule and quantity. The feeder is powered by a solar panel and a rechargeable battery, while voltage monitoring is used to observe battery condition and support reliable operation.

A smart floating boat prototype was developed to carry the fish health monitoring and water quality sensing modules. The boat uses two DC motors for navigation and four ultrasonic sensors for obstacle detection and collision prevention. A solar-powered battery system supplies energy to the onboard components, while voltage and current sensors support real-time power monitoring and priority-based power management.

Fish mortality detection is performed using a Raspberry Pi-based vision module with a camera. The module processes live video frames and applies YOLOv8-based inference to detect fish and support health condition analysis. The water quality module includes sensors for pH, turbidity, ammonia, dissolved oxygen, carbon dioxide, and temperature. Since some sensors are not suitable for continuous direct submersion, they are mounted above the water level, and a pump-based sampling mechanism is used to draw pond water into the sensing chamber.

Together, these hardware prototypes form a distributed IoT ecosystem that supports real-time sensing, automation, monitoring, alerting, and decision-making within the

proposed aquaculture system. Fig. 6 shows the IoT devices and hardware modules developed for the system implementation.

IV. RESULTS AND DISCUSSION

The proposed IoT-enabled aquaculture system was evaluated through prototype deployment and functional testing in a koi pond environment. Since the system is still in the implementation and validation stage, the evaluation focused mainly on functional correctness, module integration, response behavior, and practical usability rather than extensive statistical benchmarking. The results confirmed that the major hardware, software, communication, monitoring, and alerting functions operated as intended under prototype-level testing conditions.

The fixed automated feeding system showed stable operation during repeated feeding trials. The load-cell-based measurement mechanism supports controlled feed dispensing according to the quantity configured by the user. When the required feed quantity was reached, the dispensing mechanism stopped automatically, reducing the possibility of overfeeding. The fixed installation also improved stability because the feeding process was not affected by boat movement or navigation conditions. Manual feeding, scheduled feeding, device status monitoring, and alert handling were successfully accessed through the web dashboard. The feeding logs were recorded and made available for later cost analysis, which supports better tracking of feed usage over each farming cycle.

The smart floating boat supports mobile sensing and monitoring within the pond environment. The dual motor arrangement allowed remote movement through the dashboard, while the ultrasonic sensors helped detect obstacles and reduce the risk of collision during navigation. The power monitoring function also performed as expected during prototype testing. When battery levels decreased, the system prioritized essential functions and reduced the operation of non-essential components. This behavior showed the importance of energy-aware design in outdoor aquaculture systems, especially when solar power is used as the main energy source.

The fish mortality detection module enabled live camera-based monitoring of fish conditions using the Raspberry Pi and YOLOv8-based computer vision model. The system captured video frames from the pond environment and processed them locally to detect fish presence and support health condition analysis. In addition to image-based classification, behavior-related indicators such as low movement and abnormal floating posture were considered to improve reliability. This approach reduced dependence on single-frame decisions and supported more meaningful identification of unhealthy or dead fish conditions. Detection results were displayed on the dashboard, and alerts were generated when abnormal fish conditions were identified.

The water quality monitoring module provided continuous readings for pH, turbidity, temperature, dissolved oxygen, ammonia, and carbon dioxide. The pump-based sampling mechanism allowed sensor readings to be taken through a

sensing chamber, which was useful for sensors that were not suitable for continuous direct submersion. Rain detection was also integrated to identify weather events that could influence pond conditions and fish behavior. The dashboard allowed threshold values to be configured for each parameter. When readings exceeded the defined safe limits, the system generated alerts and maintenance reminders, supporting timely corrective action and condition-based pond management.

The budget tracking and forecasting module combined manual expense recording with IoT-based energy monitoring. Costs related to feed, labour, chemicals, maintenance, and utilities were recorded through the POS-style interface. Voltage and current readings supported electricity usage calculation, while the collected data were used to estimate production cost, cycle-based cost, and cost per surviving fish. Time-series forecasting provided preliminary future cost estimates based on available historical data. Although forecasting accuracy is expected to improve with longer data collection periods, the prototype results showed that financial analytics can be integrated effectively with aquaculture monitoring.

Overall, the results indicate that the proposed system successfully integrates feeding automation, fish mortality detection, water quality monitoring, maintenance support, energy monitoring, alerting, and cost analysis within a single web-based platform. The unified dashboard reduces the need to manage separate tools for monitoring, control, and record keeping. This integration improved operational visibility and provided farmers with timely information for decision-making. The modular design also supports future expansion, including improved disease classification, mobile application support, longer-term forecasting, and advanced decision-support functions for sustainable koi aquaculture management.

V. CONCLUSION

This research presented an IoT-enabled aquaculture management system designed for koi fish farming by integrating fixed automated feeding, real-time fish mortality detection, continuous water quality monitoring, and budget management with predictive forecasting into a unified web-based platform. The system demonstrated reliable performance through prototype implementation and functional evaluation, highlighting the effectiveness of combining IoT sensing, edge-based computer vision, automation, cloud communication, and financial analytics to improve pond management. The fixed automated feeding system supported accurate feed dispensing and reduced manual effort, while the live video monitoring module enabled early identification of unhealthy or dead fish conditions. Real-time water quality monitoring and alert generation supported proactive pond maintenance, and the budget tracking module provided useful financial insights for cost control and planning. The use of solar energy and priority-based power management further improved the suitability of the system for outdoor aquaculture environments.

Future work will focus on improving live video analytics, increasing fish health and disease classification accuracy through larger datasets, enhancing predictive control methods, and evaluating the system over longer operational periods. Further improvements may include mobile application support, advanced decision-support features, and broader multi-pond analytics to assist farmers in sustainable koi aquaculture management.

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